My research goal is to make technology safer for *every* individual. As digital technologies become increasingly entrenched and ubiquitous in everyday life, it raises privacy and safety challenges for users, especially, in control of adversarial entities. Conventional mechanisms like privacy tools or blacklisting rely on threat models that assume distant or unknown adversaries. Thus, they fail to protect victims as they overlook scenarios in which the adversary and victim are linked by a relationship, physical proximity, social identity, and power dynamic. Abusers get around these conventional security mechanisms to exploit personal technologies (e.g., stalking from an abusive partner) or structural technologies (e.g., AMBER alerts). My research studies interpersonal violence and structural violence through their underlying assertion of power and control over the victim. Stalking from an intimate partner underpins similar coercive dynamics embedded in States' use of crime databases to violate Indigenous peoples, albeit at different scales. To ensure safety, interventions must focus on the users' needs, individual and collective agency, and flourishing; moving the needle from survival to thriving.

My research examines **how technologies enables gender-based violence (GBV) such as intimate partner violence, labor/sex trafficking, and ecological extraction**. Specifically, how **equitable, scientific design interventions can detect and prevent harm among impacted communities and help them foster collective safety, resistance, and liberation**. To do this, I combine rigorous data-driven and community-engaged research methods to bridge disciplinary boundaries in and beyond Human-Computer Interaction (HCI) and Security & Privacy (S&P). I have made contributions across two woven threads:

1. **Designing interventions to support interpersonal violence survivors,** by measuring the prevalence of technology violence experiences in college context [Late-Breaking CHI 2024]. I surface barriers faced by survivors of IPV in taking supportive strategies [USENIX 2024, Late-Breaking NDSS 2024]. With GBV organizations, I have built and deployed educational interventions to embolden advocates [In-review CSCW 2027] and tools to help survivors collect legal evidence [CSCW 2025, CHI 2026, CCS 2025, In-review USENIX 2027].
2. **Research and design framework to help communities liberate from structural violence,** by demonstrating how technological infrastructures used by the State and BigTech perpetuate settler-colonial violence on Indigenous peoples [In-review CSCW 2027] and their land. By centering resistance movements, I surface a research and design framework that centers communities' resistance and reclamation of technological infrastructures, helping them liberate from structural violence.

Research Process. Globally, 1 in 3 women, 1 in 4 men, and 1 in 2 transgender/non-binary people experience physical/sexual intimate partner violence (IPV) in their lifetime [10]. Exposure to violent experiences take away power, control, and agency from survivors, making it difficult to mitigate violence and find safety. As a consequence, survivors face traumatic impacts such as, reduced confidence, difficulty in daily flourishing, cultural expression, and economics, exposure to violence down the line, health inequities, or pass on intergenerational trauma to a child. Addressing violence, therefore, requires safe, restorative, and transformative interventions that build their confidence and resilience. However, communities who face the fiercest violence have historically not been represented in academic solutions, even though, they are the experts with lived experiential knowledge. Stakeholders are often overburdened, under-compensated, and historically, taken advantage of by researchers; all of which limit their participation in research. Thus, I collaborate with community stakeholders through sensitive methods to understand their most pressing issues, structural forces that impact their technology use, and their desire for safety. I evaluate communities' needs by offering them a safe space with care and validation through a trauma-informed care and cultural sensitivity. Then, I use a diverse variety of methods to collect data through interviews/focus groups, surveys, large-scale content analysis, autoethnography, and system design, and used mixed-methods approaches to analyze them. Finally, I utilize "Upstream Story" framework (see Figure 1) to build interventions that can (a) fulfill communities' immediate needs and transform violence towards justice and (b) reduce harm by enabling stakeholders to detect violence early. Then, I work with the communities to iterate on the intervention design, assess its effectiveness in practice. Overall, this process is extremely challenging: emotionally, logistically, and economically complex with high-stakes impact on stakeholders' life, especially with political uncertainties related to identity-focused research. It requires a thorough understanding across disciplines spanning system design, healthcare, anthropology, law, politics, history, education, and, implementation science.

1. Support Interpersonal Violence Survivors. To direct interventional research and resources effectively, it is crucial to understand the statistics of the most and least frequent tech violence experiences and support-seeking strategies. Using college campuses as one such context, I studied [Late-Breaking CHI 2024]

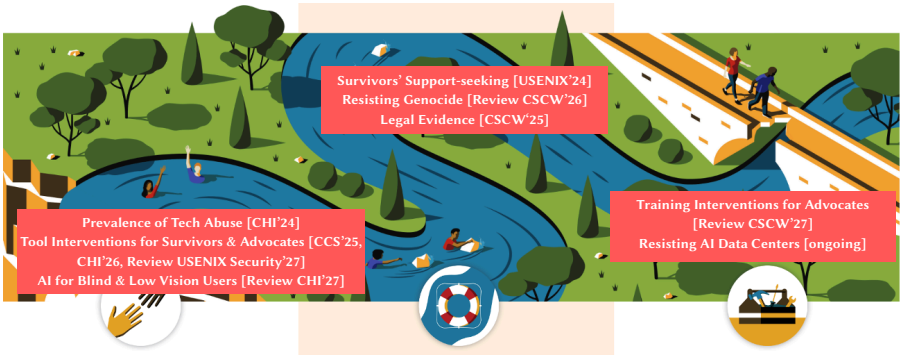


Figure 1: My work focuses on each part of the stream to effectively prevent violence by building interventions (a) downstream to support survivors' immediate needs after the abuse has occurred, (b) midstream to support survivors' coping strategies to mitigate abuse, and (c) upstream to prevent survivors from falling into the river.

the prevalence of interpersonal tech violence and support strategies, through a survey of 776 college students. Alarmingly, 70% of students experienced such experiences: 60% faced unwanted calls and messages, 51% non-consensual sharing of intimate images/videos or messages. Also, women (2x) and LGBTQ+ (3x) students were more likely to face violence than men, showing that tech violence is a gendered issue. Importantly, more than half of students (53%) did not seek support. When they did, most students (93%) solely relied on informal systems like friends and family, and only 6% solely sought formal systems. Therefore, an early intervention may nudge survivors to reach out to informal systems.

The prevalence study, however, does not explain *why* survivors do not seek support and what happens when they do. To understand this, I interviewed nine survivors to examine their experiences in seeking support from their community and the effectiveness of support in addressing their needs. This study [USENIX 2024] underlined the barriers that impeded the survivors from seeking support and the challenges during the support process. Survivors and support systems downplay the severity of violence and deem it as lower priority than physical or sexual violence due to a lack of visible signs of abuse. Survivors are often disbelieved and provided with an ineffective advocate to mitigate abuse due to a lack of adequate resources. Moreover, support providers use unsafe technological mediums to communicate, which is especially challenging when abusers often isolate them or escalate violence. I discovered an ecosystem of care by survivors who participated in and exchanged help to other survivors through social media support groups. The key takeaway is that, despite not being an expert, survivors and their informal systems collaborated to improve their healing and agency for recovery.

To foster this ecosystem of care, my next study [In-review CSCW 2027] conducted educational interventions to build knowledge capacity among advocates in collaboration with eight host GBV organizations. Advocates face knowledge gaps in identifying and mitigating newer, ever-evolving forms of tech violence, which becomes a deterrent for them in providing effective support to survivors. Through pre- and post-surveys, we found that the trainings increased advocates' perceived understanding of vectors of violence and their impacts on survivors. Trainings increased advocates' confidence in providing support, safety planning, and referrals through collaboration with community stakeholders. Then, I conducted retrospective reflections through autoethnography to surface key lessons, contestations, and tensions that emerged in designing and facilitating trainings. To help scholars design their own interventions in respective high-stakes contexts, the study creates a concrete checklist based on key phases of the study: coordinating logistics flexibly, reducing reflexive distance from host organizations and attendees, and using field experience to tailor content for accessibility. Moreover, I helped design *Sherloc* [CSCW 2025, CHI 2026], a tool that helps consultants capture and prepare an evidence report that can help survivor get relief legal injunctions (restraining order/divorce settlement). Similarly, *Lestrade* [CCS 2025, In-review USENIX 2027], reduces uncertainty within account security interfaces by utilizing account data exports to create a visual timeline to aide clinical investigation.

2. Help Communities Liberate from Settler-Colonial Violence. This line of work critiques the historical and ongoing role of technologies in perpetuating structural violence. Through this work, I expand the how violence and trauma can manifest in and beyond bodies, on ecology, and knowledge systems. On the other hand oppressed communities reconfigure technologies for advocacy, healing, and self-determination. This thread envisions a framework that critically engages with epistemic injustices and settler-colonial logics of extraction and domination, while, centering resistance and self-determination in research and design of interventions that

liberate communities from structural violence.

In the US, despite less than 3% of the population, 84.3% of Indigenous women face violence from a partner when 97% abusive partners are non-native [11]. Indigenous women are 10x more likely to be murdered than the national average [12] and represent 40% of sex trafficking victims [13]. However, most tribal nations cannot prosecute non-Native perpetrators due to lack of jurisdictional sovereignty, giving rise to the ongoing genocide or the Missing & Murdered Indigenous Relatives (MMIR) crisis. Comprehensive data on the exact number remains scarce in federal criminal databases, leaving most cases unresolved or inadequately investigated. Lucchesi and Echo-Hawk [14] found 5,712 cases of missing & murdered people but only 116 (2%) of them were recorded in the US databases. I analyzed 140 news articles, advocacy & police reports, podcasts, and court hearings that consisted of stories of missing relatives, families, advocates, and tribal police. I found that inaccurate data collection, opaque data sharing policies, and lack of safe online spaces create challenges for communities to find relatives. Further, colonial institutions have historically misused and appropriated Indigenous knowledge without consultation. For example, the Royal Canadian Mountain Police installed five 5G towers along the "The Highway of Tears" to improve spotty cell service. However, without consulting Indigenous leadership, they left out many highways with spotty cell coverage by using a narrow definition of missing or murdered. Right now, most Tribes lack the access to the State criminal databases and agency to release alerts, forcing dependence on colonial institutions. Notably, Indigenous communities mount a generational resistance against State technologies and reclaim them as spaces of advocacy, memory, and care. Communities form a care ecosystem through missing-poster campaigns on social media, sovereign and crowd-sourced databases, and digital maps to search and rescue for relatives, provide support and healing to families, and raise awareness of the MMIR crisis. Recognizing the impact, UMichigan Prof. Rebecca Irvine formally invited me to collaborate on a book.

AI has been touted as deeply transformational, having the potential to reshape our society on a global scale, fueling an accelerationist push towards scientific progress. However, under the false promises of jobs, economic, and renewable energy development, BigTech companies [1] are scaling out hyperscale AI Data Centers (DC) which extract from Black and Indigenous lands. For example, in 2023, Google's DC used over 66 billion litters of water for cooling, surpassing the water use of Los Angeles for 2 weeks [15]. Through this project, we counter the dominant AI accelerationist narrative by critically understanding the physical impacts of violence caused by DC on Black and Indigenous communities. To do this, I participated in webinars hosted HonorTheEarth and No Desert Data Center Coalition to conduct ethnographic interviews with the organizers. However, organizers highlighted the urgency of the movement which is more pressing than academic research. Respecting their refusal, I instead scraped online articles, reports, podcasts that held stories about impacts of violence on communities. Our initial findings show that DC colonize intersecting embodied and ecological connections: (a) economic extraction with inflated energy bills, job losses due to automation, (b) health & climatic conditions: air and noise pollution, rise in cancer rates, climate catastrophe (c) direct violence: severing relational connections with nature, coercion, harassment, and trafficking, and (d) agency and epistemology: forcing capitalist visions of the future. To fight back, communities are leading resistance movements that (a) refuse the privatization of their land and (b) reclaim technologies to organize knowledge-sharing workshops, responsible land-use policies, and social media activism. This work re-imagines how sovereign and culturally-sensitive, sustainable, sovereign, and ethical AI can manifest and benefit the communities.

Future Agenda. In the coming years, I will continue to engage with communities who face violence caused by rapid development of new and emerging technologies (e.g., LLMs, wearables, federal alerts, AI nudification apps, robots). To inform safety interventions, I will setup a community-engaged research program through a Tech Clinic in Copenhagen. In short-term, I will develop interventions that helps stakeholders detect and prevent interpersonal violence early and more effectively. In parallel, I will research how emerging technologies accelerate structural violence towards marginalized communities. And in long-term, I will theorize and build safe, ethical, responsible, and sovereign technologies that dismantle structures of violence.

1. Early Interventions to Empower IPV Stakeholders. Tech Clinics are a scarce resource that is not accessible to most survivors outside four metropolitan US cities [2]. One direction will **(a) enable advocates in healthcare clinics to detect interpersonal tech violence early**. Survivors typically fill out intake forms when they enter a hospital or shelter. Intake forms help first-responders (emergency rooms & sexual assault nurses) assess psychological, economic, and physiological safety. Embedding tech violence experiences questionnaires into

[1] Big Tech companies include OpenAI, Anthropic, Google, Meta, xAI.

[2] There are four tech clinics in the US: CETA in New York City, MTC in Madison, TECC in Seattle, and MAVS-ETA in Arlington.

the intake forms will enable providers to assess their digital safety early that can reduce trauma and health impacts by flagging and redirecting them to appropriate support providers in time. I have co-developed questions to help educate first-responders identify signs of tech violence in their daily roll-call trainings. Secondly, I plan to signal a shift from individual safety plans towards a collectivist notion of safety, I will **(b) scale GBV providers to a peer support model** where healing happens with the community. In this model, peer support workers (PSWs) run support groups by sharing stories of violence & resilience [16]. These groups create a destigmatized space that redistributes epistemic and decision-making power, build technical skills, learn folk threat models, making safety planning less daunting for survivors. Utilizing my experience in training advocates [In-review CSCW 2027], I plan to co-design a structured program that empowers PSWs to effectively moderate and facilitate support groups. Finally, for rural and BIPOC survivors, culturally sensitive support systems are often inaccessible. Geographic isolation leads to access barriers: lack of internet, digital literacy, resource constraints, and social barriers: cultural stigma, a close-knit community, police siding with abusers, slow justice system response. I will build a **(c) culturally-attuned tech clinic** that weighs survivors' dynamically changing cultural safety needs. This intervention will leverage the embedded knowledge infrastructure, weaving technology astuteness among the youth and generational teachings from Indigenous Elders. At MTC, we have supported 40 survivors with 15 rural organizations so far, by integrating remote consultations over Zoom in our service delivery protocol. My ongoing work evaluates the effectiveness of such an intervention model and how tech clinic consultants have navigated these consultations.

2. Surfacing and Resisting Structural Violence. My work will continue to reckon with structural violence faced by communities in the Majority World (e.g., war refugees, incarcerated, factory & delivery workers, immigrants, unhoused, construction workers). Intergenerational trauma faced by these communities have lasting health impacts. Such negative impacts may complicate their privacy & safety needs as more essential sector technologies get digitized. For example, what does privacy mean for international students who go through social media surveillance by the State to maintain their immigration status? Similarly, what does privacy mean for exploited factory workers in India who watch abusive content or wear cameras to train AI robots that will soon replace them? I plan to answer these questions through intersectional theoretical lenses; such as queer [17] and black feminism [18], caste oppression [19], anti-incarceration and abolition [20, 21], labor exploitation and surveillance capitalism [22, 23], and refugee resettlement [24]. My projects will **(a) measure the impacts of privacy challenges faced by communities subjected to structural violence**. Importantly, to acknowledge their daily flourishing and resistance, I will examine how **(b) communities use social media spaces for connection, organizing, and justice movements** and how their lived experiences can inform interventions and tech policy. Recognizing culture as a vital safety condition, my work will design **(c) frameworks for cultural preservation and revitalization through technologies** through stories, knowledge, languages at-risk of extinction. Finally, collaborating with health epidemiologists and violence-prevention practitioners, I will **(d) design technological visualizations tracing the systemic health inequities** with an aim towards providing culturally-competent interventions to curb them.

3. Decolonizing AI Safety Research. Scholars have noted that computing research and interventions reflect an imperial impulse [25–27]. However, without engaging with theories of change that center decoloniality, sovereignty, and liberation [28, 29], safety research may result in neoliberal siloed interventions that fail to address the privacy violations of peoples that power the Big Tech enterprise. For example, trust & safety teams in Big Tech companies often tout their products as private, ethical, carbon-neutral, whereas, the same companies actively harm miners who extract rare earth minerals in Congo and assemble their products in India. My computing and critical theory expertise allows me to view technologies not just as a black box, but as a combination of complex social and technical components. I will provide evidence to decolonize the definition of safety & privacy by tracing privacy violations in the entire lifecycle of BigTech AI products. To do this, I will conduct a **(a) review the positionality of scholarships that underpin the digital safety research**. Is the prior effectively addressing a variety of diverse harms and safety needs across AI abstraction layers, or, reinforcing a limited definition of online safety? Then, drawing on my workshops [Workshop HLF 2026, Proposal CSCW 2024, In-review Workshop CHI 2027, In-review Workshop DIS 2026], I plan to **(b) create a pluriversal framework for AI safety** by expanding the visions of privacy and impacts of violence to situate safety in Global South and North. I will contrast how AI proliferation creates conditions of danger within communities due to the fundamental value misalignment between the corporations and communities. The framework will provide opportunities to shape AI towards a safe, sustainable, sovereign, and ethical AI.

My Publications

[Late-Breaking CHI 2024] **Naman Gupta**, Sanchari Das, Kate Walsh, and Rahul Chatterjee. "A Critical Analysis of the Prevalence of Technology-Facilitated Abuse in US College Students". In: *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems*. 2024. URL: https://doi.org/10.1145/3613905.3652036.

[USENIX 2024] **Naman Gupta**, Kate Walsh, Sanchari Das, and Rahul Chatterjee. ""I really just leaned on my community for support": Barriers, Challenges, and Coping Mechanisms Used by Survivors of {Technology-Facilitated} Abuse to Seek Social Support". In: 2024. URL: https://dl.acm.org/doi/10.5555/3698900.3699179.

[Late-Breaking NDSS 2024] **Naman Gupta**, Rose Ceccio, Majed Almansoori, and Rahul Chatterjee. "Analyzing the Patterns and Behavior of Users When Detecting and Preventing Tech-enabled Stalking". In: 2023. URL: https://doi.org/10/gr8rnx.

[In-review CSCW 2027] **Naman Gupta**, Connie W. Chau, Sophie Stephenson, Rahul Chatterjee, and Kate Walsh. "Lessons from Trauma-Informed Training on Technology-Facilitated Abuse for Gender-Based Violence Advocates". In: *review at CSCW* (2026). URL: https://arxiv.org/abs/2610.02688.

[CSCW 2025] Sophie Stephenson, **Naman Gupta**, Akhil Polamarasetty, Kyle Huang, David Youssef, Kayleigh Cowan, and Rahul Chatterjee. "Legal Evidence of Technology-Facilitated Abuse in Wisconsin: Surfacing Barriers Within and Beyond the Courtroom." In: *Computer Supported Cooperative Work and Social Computing* 🏆 **Honorable Mention (top 5%)**. 2025. URL: https://doi.org/10.1145/3757622.

[CHI 2026] Sophie Stephenson, **Naman Gupta**, Kyle Huang, David Youssef, Kayleigh Cowan, and Rahul Chatterjee. "Trauma-Informed Digital Evidence Collection:A Design Inquiry into Evidence Practices for Technology-Facilitated Abuse in Intimate Partner Violence". In: *2026 CHI Conference on Human Factors in Computing Systems* 🏆 **Best Paper (top 1%)**. 2026. URL: https://doi.org/10.1145/3772318.3790296.

[CCS 2025] Julia Nonnenkamp, **Naman Gupta**, Abhimanyu Dev Gupta, and Rahul Chatterjee. "Hidden in Plain Bytes: Investigating Interpersonal Account Compromise with Data Exports". In: *2025 ACM SIGSAC Conference on Computer and Communications Security*. 2025. URL: https://doi.org/10.1145/3719027.3765147.

[In-review USENIX 2027] Julia Nonnenkamp, **Naman Gupta**, Angela Wang, Sophie Stephenson, and Rahul Chatterjee. "LEStrADE: Investigating Account Compromise using Data Exports for Survivors of Technology-Facilitated Abuse". In: *review at USENIX Security*. 2026. URL: https://naman.github.io/assets/files/papers/lestrade.pdf.

[In-review CSCW 2027] **Naman Gupta**, Sophie Stephenson, Chung Chi Yeung, Wei Ting Wu, Jeneile Luebke, Kate Walsh, and Rahul Chatterjee. ""Lighting The Way For Those Not Here": How Technology Researchers Can Help Fight the Missing and Murdered Indigenous Relatives (MMIR) Crisis". In: *review at CSCW 2027*. 2025. URL: https://doi.org/10.48550/arXiv.2601.17966.

[Workshop HLF 2026] **Naman Gupta**. "Polycrisis in the Age of AI and Scientific Acceleration". In: 2026.

[Proposal CSCW 2024] **Naman Gupta** and Rahul Chatterjee. "Decolonizing Digital Safety Research: Safety for Who? Safety from What?" In: *CSCW 2024* (2024).

[In-review Workshop CHI 2027] **Naman Gupta**, Freddy Reiber, Jasmine C. Foriest, Ailsa Robertson, Jordan Taylor, Palak Dudani, Ghadeer Awwad, Shreya Chowdhary, David Gamba, Eugene Kim, and Katie Seaborn. "Counter-Computing: Digital Resistances and Global Solidarities". In: *Review at ACM CHI Conference on Human Factors in Computing Systems*. 2027.

[In-review Workshop DIS 2026] Jasmine C Foriest, Alicia DeVrio, Li Qiwei, Nimra Ahmed, **Naman Gupta**, Jensine Raihan, Franchesca Spektor, and Justine Zhang. "Violence as a Boundary Object: Language, Power, and Design Justice". In: *Review at ACM DIS Designing Interactive Systems*. 2026.

References

[10] WHO. *Devastatingly pervasive: 1 in 3 women globally experience violence*. 2021.

[11] Andrè B Rosay. "Violence Against American Indian and Alaska Native Women and Men". In: *National Institute of Justice* (2016).

[12] Ronet Bachman, Heather Zaykowski, Rachel Kallmyer, Margarita Poteyeva, and Christina Lanier. "Violence Against American Indian and Alaska Native Women and the Criminal Justice Response: What is Known". In: (2008).

[13] National Congress of American Indians Policy Research Center. *Human & Sex Trafficking: Trends and Responses Across Indian Country*. Tech. rep. 2016.

[14] Annita Lucchesi and Abigail Echo-Hawk. *Missing and Murdered Indigenous Women & Girls: A snapshot of data from 71 urban cities in the United States*. Tech. rep. 2018.

[15] Nuoa Lei, Jun Lu, Arman Shehabi, and Eric Masanet. "The water use of data center workloads: A review and assessment of key determinants". In: *Resources, Conservation and Recycling* (2025).

[16] Max Osborn, Tanyanne Ball, and Valli Rajah. "Peer Support Work in the Context of Intimate Partner Violence: A Scoping Review". In: *Trauma, Violence, & Abuse* (2024).

[17] Gloria Anzaldúa. *Borderlands =: La frontera: the new mestiza*. Fifth edition. Aunt Lute Books, 2021.

[18] Patricia Hill Collins. *Black Feminist Thought in the Matrix of Domination*. 1990.

[19] Palashi Vaghela, Steven J Jackson, and Phoebe Sengers. "Interrupting Merit, Subverting Legibility: Navigating Caste In 'Casteless' Worlds of Computing". In: *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. CHI '22. Association for Computing Machinery, 2022.

[20] Anna Terwiel. "What Is Carceral Feminism?" In: *Political Theory* (2020).

[21] Ruha Benjamin. *Captivating Technology: Race, Carceral Technoscience, and Liberatory Imagination in Everyday Life*. Duke University Press, 2019.

[22] Michael Hardt and Antonio Negri. *Empire*. 1. Harvard Univ. Press paperback ed., [Nachdr.] Harvard Univ. Press, 2003.

[23] Shoshana Zuboff. "The Age of Surveillance Capitalism". In: *Social Theory Re-Wired*. 3rd ed. Routledge, 2019.

[24] Sarah Nandi. "Making Gender-Based Violence Count: Objectivity, Expertise, and Humanitarian Data Collection in the Refugee Regime". In: *Global Stud. Q.* (2026).

[25] Paul Dourish and Scott D. Mainwaring. "Ubicomp's colonial impulse". In: *Proceedings of the 2012 ACM Conference on Ubiquitous Computing*. UbiComp '12. Association for Computing Machinery, 2012.

[26] Syed Mustafa Ali. "A Brief Introduction to Decolonial Computing". In: *XRDS* (2016).

[27] Jensine Raihan. "The Colonial Legacy of the Human in Human-Computer Interaction: Implications for Critical Design Approaches". In: *Proc. Sixth Decenn. Aarhus Conf. Comput. X Crisis*. AAR '25. Association for Computing Machinery, 2025.

[28] Arturo Escobar. *Designs for the Pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds*. Duke University Press, 2018.

[29] Morgan Ndlovu. "Decoloniality Is Agency". In: (2025).